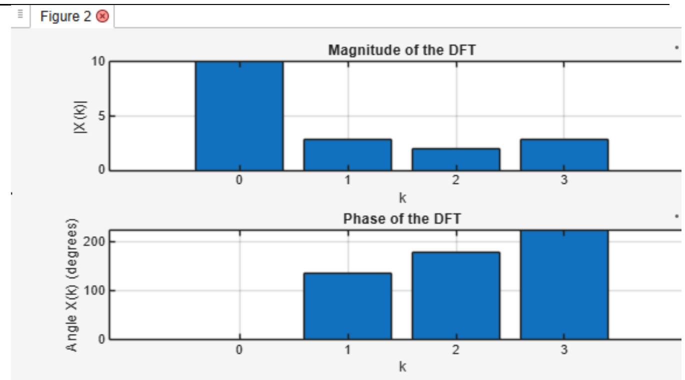
9. DFT and FFT Algorithms

Experiment 9.1

Discrete Fourier Transform

```matlab
% Test Case: Discrete Fourier Transform (DFT) of a Discrete-Time Signal

% Define the discrete time signal
x_n = [1 2 3 4];         % Signal x[n]

% Compute the Discrete Fourier Transform (DFT)
X_k = fft(x_n);

% Define the frequency axis
N = length(x_n);         % Number of points in the DFT
k = 0:N-1;               % Frequency index

% Compute magnitude of DFT
magnitude = abs(X_k);

% Initialize phase array
phase = zeros(1, N);

% Compute phase manually
for i = 1:N
    real_part = real(X_k(i));
    imag_part = imag(X_k(i));

    if real_part > 0
        phase(i) = atan(imag_part / real_part);

    elseif real_part < 0
        phase(i) = atan(imag_part / real_part) + pi;

    else
        if imag_part > 0
            phase(i) = pi/2;

        elseif imag_part < 0
            phase(i) = -pi/2;

        else
            phase(i) = 0;
        end
    end
end
```

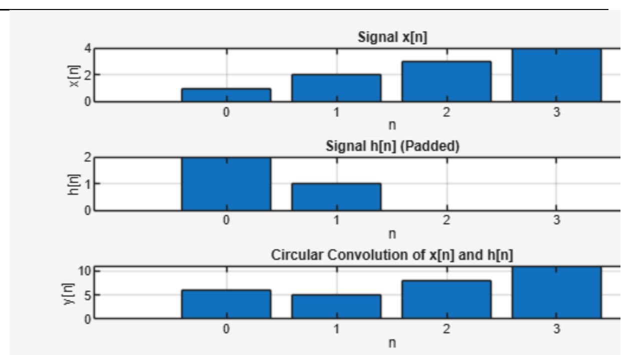


```
% Convert phase from radians to
degrees
phase_deg = phase * 180/pi;

% Plot the magnitude and phase of the
DFT
figure;

% Plot magnitude
subplot(2,1,1);
bar(k, magnitude);
xlabel('k');
ylabel('|X(k)|');
title('Magnitude of the DFT');
grid on;

% Plot phase
subplot(2,1,2);
bar(k, phase_deg);
xlabel('k');
ylabel('Angle X(k) (degrees)');
title('Phase of the DFT');
grid on;
```

Experiment 9.2

Circular Convolution

```
% Test Case: Circular Convolution of
Discrete-Time Signals

% Define the discrete time signals
x_n = [1 2 3 4];        % Signal x[n]
h_n = [2 1];            % Signal h[n]

% Zero-pad h[n] to match the length of
x[n]
N = length(x_n);        % Length of signal
x[n]
h_n_padded = [h_n zeros(1, N -
length(h_n))];

% Compute the circular convolution
y_n = cconv(x_n, h_n_padded, N);

% Define the time index for the output
signal
n_y = 0:N-1;

% Plot the input signals and their
circular convolution
```



<pre>figure;
 	
% Plot x[n]
subplot(3,1,1);
bar(0:length(x_n)-1, x_n);
xlabel('n');
ylabel('x[n]');
title('Signal x[n]');
grid on;
 	
% Plot h[n] (padded)
subplot(3,1,2);
bar(0:N-1, h_n_padded);
xlabel('n');
ylabel('h[n]');
title('Signal h[n] (Padded)');
grid on;
 	
% Plot y[n] (circular convolution result)
subplot(3,1,3);
bar(n_y, y_n);
xlabel('n');
ylabel('y[n]');
title('Circular Convolution of x[n] and
h[n]');
grid on;</pre>	